```
> # PROOF OF IDENTITIES FOR a_7(n)
  with(qseries):
  with(ETA):
  t := gp2etaprod([7, 4, 1, -4]):
  p0 := gp2etaprod([14, 4, 1, 4, 7, -4, 2, -4]):
  p1 := (gp2etaprod([14, 1, 1, 7, 7, -1, 2, -7]) - 1)/7:
  A7 := gp2etaprod([2, 6, 49, 7, 1, -7, 98, -6]):
  B7 := 1/q:

> # GROUP 1, IDENTITY 1: U_A7(p0)
  EP7_1_1 := normal(A7*p0):
  RHS7_1_1 := expand(p0*(7^8*t^5 + 34*7^6*t^4 + 405*7^4*t^3 + 2150*7^2*t^2
      + 688*7*t + 8*7) + p1*(-7^7*t^4 - 33*7^5*t^3 - 381*7^3*t^2 - 40*7^3*t
      - 87*7 - 7*t^(-1))) + 9*7^6*t^4 + 255*7^4*t^3 + 393*7^3*t^2 + 1968*7*t
      + 87*7 + 7*t^(-1)):
  proof7_1_1 := provemodfuncGAMMA0UpETAidBATCH(EP7_1_1, 7, RHS7_1_1, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 17    This means checking up to q^(18).
We find that LHS - RHS is
```

$$O(q^{45})$$

$$proof7_1_1 := [1, -17, O(q^{45})] \quad \textbf{(1)}$$

```
> # GROUP 1, IDENTITY 2: U_A7(p0*t^(-1))
  EP7_1_2 := normal(A7*p0/t):
  RHS7_1_2 := expand(p0*(7^3*t^2 + 24*7*t + 16) + p1*(-7^2*t - 23 - 2*t^(-1))
      + 9*7*t + 23 + 2*t^(-1)):
  proof7_1_2 := provemodfuncGAMMA0UpETAidBATCH(EP7_1_2, 7, RHS7_1_2, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 8    This means checking up to q^(9).
```

We find that LHS - RHS is

$$O(q^{36})$$

$$proof7_1_2 := [1, -8, O(q^{36})] \quad \textbf{(2)}$$

```
> # GROUP 1, IDENTITY 3: U_A7(p0*t^(-2))
  EP7_1_3 := normal(A7*p0/t^2):
  RHS7_1_3 := expand(p0*(-7^5*t^3-44*7^3*t^2-464*7*t-208)+p1*(7^4*t^2
      +43*7^2*t+432+26*t^(-1))-11*7^3*t^2-47*7^2*t-432-26*t^(-1)):
  proof7_1_3 := provemodfuncGAMMA0UpETAidBATCH(EP7_1_3, 7, RHS7_1_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 11    This means checking up to q^(12).
We find that LHS - RHS is
```

$$O(q^{39})$$

$$proof7_1_3 := [1, -11, O(q^{39})] \quad \textbf{(3)}$$

```
> # GROUP 1, IDENTITY 4: U_A7(p0*t^(-3))
  EP7_1_4 := normal(A7*p0/t^3):
  RHS7_1_4 := expand(p0*(-7^7*t^4-32*7^5*t^3-54*7^3*t^2+2096*7*t+192
      *7)+p1*(7^6*t^3+31*7^4*t^2+34*7^2*t-286*7-24*7*t^(-1))-11*7
      ^5*t^3-209*7^3*t^2-8*7^2*t+286*7+24*7*t^(-1)):
  proof7_1_4 := provemodfuncGAMMA0UpETAidBATCH(EP7_1_4, 7, RHS7_1_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 14    This means checking up to q^(15).
We find that LHS - RHS is
```

$$O(q^{42})$$

$$proof7_1_4 := [1, -14, O(q^{42})] \quad \textbf{(4)}$$

```
> # GROUP 1, IDENTITY 5: U_A7(p0*t^(-4))
```

```
EP7_1_5 := normal(A7*p0/t^4):
RHS7_1_5 := expand(p0*(26*7^7*t^4+741*7^5*t^3+5034*7^3*t^2+353*7^2*t
    -6432)+p1*(-26*7^6*t^3-717*7^4*t^2-652*7^3*t-194*7+804*t^(
    -1))+2*7^7*t^4+262*7^5*t^3+107*7^5*t^2+4440*7^2*t+194*7-804
    *t^(-1))):
proof7_1_5 := provemodfuncGAMMA0UpETAidBATCH(EP7_1_5,7,RHS7_1_5,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$O(q^{44})$$

$$proof7_1_5 := [1, -16, O(q^{44})] \quad \textbf{(5)}$$

```
> # GROUP 1, IDENTITY 6: U_A7(p0*t^(-5))

    # Multiply both sides by t^2 to remove the negative powers, using U_7(t(7*tau)^2*G)=t^2*
    U_7(G).
EP7_1_6 := normal(gp2etaprod([49,8,7,-8])*A7*p0/t^5):
RHS7_1_6 := expand(t^2*(p0*(7^11*t^6+26*7^9*t^5-94*7^7*t^4-1170*7^6*t^3
    -10022*7^4*t^2-22922*7^2*t+16152+8*t^(-1))+p1*(-7^10*t^5-25*7
    ^8*t^4+16*7^7*t^3+163*7^6*t^2+9220*7^3*t+19769*7-2020*t^(-1)-t
    ^(-2))+7^10*t^5+139*7^7*t^4-36*7^7*t^3-8539*7^4*t^2-64240*7^2*t
    -138375+2020*t^(-1)+t^(-2))):
proof7_1_6 := provemodfuncGAMMA0UpETAidBATCH(EP7_1_6,7,RHS7_1_6,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 24    This means checking up to q^(25).
We find that LHS - RHS is
```

$$O(q^{52})$$

$$proof7_1_6 := [1, -24, O(q^{52})] \quad \textbf{(6)}$$

```
> # GROUP 1, IDENTITY 7: U_A7(p0*t^(-6))
  EP7_1_7 := normal(A7*p0/t^6) :
  RHS7_1_7 := expand(p0*(7^13*t^7+32*7^11*t^6+445*7^9*t^5+7342*7^7*t^4
      +16095*7^6*t^3+115958*7^4*t^2+315003*7^2*t+109656-24*7*t^(-1))
      +p1*(-7^12*t^6-31*7^10*t^5-425*7^8*t^4-1019*7^7*t^3-15420*7^5*t
      ^2-106814*7^3*t-277603*7-13686*t^(-1)+3*7*t^(-2)))+11*7^11*t^6
      +297*7^9*t^5+571*7^8*t^4+8781*7^6*t^3+114308*7^4*t^2+749998*7
      ^2*t+277579*7+13686*t^(-1)-3*7*t^(-2)) :
  proof7_1_7 := provemodfuncGAMMA0UpETAidBATCH(EP7_1_7, 7, RHS7_1_7, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 25    This means checking up to q^(26).
We find that LHS - RHS is
```

$$O(q^{53})$$

$$proof7_1_7 := [1, -25, O(q^{53})] \qquad (7)$$

```
> # GROUP 2, IDENTITY 1: U_A7(p1)
  EP7_2_1 := expand(A7*p1) :
  RHS7_2_1 := expand(p0*(-6*7^13*t^8-212*7^11*t^7-3198*7^9*t^6-3788*7^8
      *t^5-18486*7^6*t^4-52512*7^4*t^3-79404*7^2*t^2-7152*7*t-16*7)+p1
      *(6*7^12*t^7+207*7^10*t^6+3053*7^8*t^5+3540*7^7*t^4+16935*7^5*t
      ^3+47245*7^3*t^2+70359*7*t+128*7^2+2*7*t^(-1)))-7^13*t^8-10*7
      ^12*t^7-1773*7^9*t^6-3333*7^8*t^5-25701*7^6*t^4-120081*7^4*t^3
      -331735*7^2*t^2-70375*7*t-128*7^2-2*7*t^(-1)) :
  proof7_2_1 := provemodfuncGAMMA0UpETAidBATCH(EP7_2_1, 7, RHS7_2_1, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 27    This means checking up to q^(28).
We find that LHS - RHS is
```

$$O(q^{55})$$

$$proof7_2_1 := [1, -27, O(q^{55})] \quad (8)$$

```
> # GROUP 2, IDENTITY 2: U_A7(p1*t^(-1))
  EP7_2_2 := expand(A7*p1/t):
  RHS7_2_2 := expand(p0*(-8*7*t-8)+p1*(8+t^(-1))-7*t-8-t^(-1)):
  proof7_2_2 := provemodfuncGAMMA0UpETAidBATCH(EP7_2_2, 7, RHS7_2_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$O(q^{34})$$

$$proof7_2_2 := [1, -6, O(q^{34})] \quad (9)$$

```
> # GROUP 2, IDENTITY 3: U_A7(p1*t^(-2))
  EP7_2_3 := expand(A7*p1/t^2):
  RHS7_2_3 := expand(p0*(8*7^3*t^2+120*7*t+64)+p1*(-8*7^2*t-113-8*t
      ^(-1))+7^3*t^2+64*7*t+113+8*t^(-1)):
  proof7_2_3 := provemodfuncGAMMA0UpETAidBATCH(EP7_2_3, 7, RHS7_2_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$O(q^{37})$$

$$proof7_2_3 := [1, -9, O(q^{37})] \quad (10)$$

```
> # GROUP 2, IDENTITY 4: U_A7(p1*t^(-3))
  EP7_2_4 := expand(A7*p1/t^3):
  RHS7_2_4 := expand(p0*(4*7^5*t^3-48*7^3*t^2-160*7^2*t-624)+p1*(-4*7
      ^4*t^2+51*7^2*t+1058+78*t^(-1))+7^5*t^3+20*7^3*t^2-439*7*t
      -1058-78*t^(-1)):
  proof7_2_4 := provemodfuncGAMMA0UpETAidBATCH(EP7_2_4, 7, RHS7_2_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 12    This means checking up to q^(13).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{40})$$

$$proof7_2_4 := [1, -12, \mathrm{O}(q^{40})] \quad \textbf{(11)}$$

```
> # GROUP 2, IDENTITY 5: U_A7(p1*t^(-4))
  EP7_2_5 := expand(A7*p1/t^4):
  RHS7_2_5 := expand(p0*(-8*7^7*t^4-248*7^5*t^3-1528*7^3*t^2+288*7^2*t
      +544*7)+p1*(8*7^6*t^3+241*7^4*t^2+1368*7^2*t-320*7-68*7*t^(
      -1))-7^7*t^4-88*7^5*t^3-249*7^4*t^2-1296*7^2*t+320*7+68*7*t^(
      -1)):
  proof7_2_5 := provemodfuncGAMMA0UpETAidBATCH(EP7_2_5, 7, RHS7_2_5, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{43})$$

$$proof7_2_5 := [1, -15, \mathrm{O}(q^{43})] \quad \textbf{(12)}$$

```
> # GROUP 2, IDENTITY 6: U_A7(p1*t^(-5))

      # Multiply both sides by t^2 to remove the negative powers, using U_7(t(7*tau)^2*G)=t^2*
      U_7(G).
  EP7_2_6 := expand(gp2etaprod([49, 8, 7, -8])*A7*p1/t^5):
  RHS7_2_6 := expand(t^2*(p0*(6*7^9*t^5+292*7^7*t^4+762*7^6*t^3+5116*7
      ^4*t^2+9322*7^2*t-14712-8*t^(-1))+p1*(-6*7^8*t^4-41*7^7*t^3
      -731*7^5*t^2-4680*7^3*t-7895*7+1840*t^(-1)+t^(-2))+7^9*t^5+78
      *7^7*t^4+367*7^6*t^3+5373*7^4*t^2+32488*7^2*t+55257-1840*t^(
      -1)-t^(-2))):
```

```
proof7_2_6 := provemodfuncGAMMA0UpETAidBATCH(EP7_2_6, 7, RHS7_2_6, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 22    This means checking up to q^(23).
We find that LHS - RHS is
```

$$O(q^{50})$$

$$proof7_2_6 := [1, -22, O(q^{50})] \quad \textbf{(13)}$$

```
> # GROUP 2, IDENTITY 7: U_A7(p1*t^(-6))

    # Multiply both sides by t^2 to remove the negative powers, using U_7(t(7*tau)^2*G)=t^2*
    U_7(G).
  EP7_2_7 := expand(gp2etaprod([49, 8, 7, -8]) * A7 * p1 / t^6) :
  RHS7_2_7 := expand(t^2 * (p0 * (-4 * 7^11 * t^6 - 232 * 7^9 * t^5 - 5924 * 7^7 * t^4 - 11280
      * 7^6 * t^3 - 70204 * 7^4 * t^2 - 166408 * 7^2 * t - 12528 + 160 * t^(-1)) + p1 * (4 * 7
      ^10 * t^5 + 229 * 7^8 * t^4 + 822 * 7^7 * t^3 + 1532 * 7^6 * t^2 + 64370 * 7^3 * t
      + 145850 * 7 + 1546 * t^(-1) - 20 * t^(-2)) - 7^11 * t^6 - 68 * 7^9 * t^5 - 2323 * 7^7
      * t^4 - 6810 * 7^6 * t^3 - 78618 * 7^4 * t^2 - 450958 * 7^2 * t - 1020790 - 1546 * t^(-1)
      + 20 * t^(-2))) :
  proof7_2_7 := provemodfuncGAMMA0UpETAidBATCH(EP7_2_7, 7, RHS7_2_7, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 25    This means checking up to q^(26).
We find that LHS - RHS is
```

$$O(q^{53})$$

$$proof7_2_7 := [1, -25, O(q^{53})] \quad \textbf{(14)}$$

```
> # GROUP 3, IDENTITY 1: U_A7(1)
  EP7_3_1 := normal(A7) :
  RHS7_3_1 := expand(p0 * (6 * 7^14 * t^8 + 212 * 7^12 * t^7 + 3198 * 7^10 * t^6 + 3788 * 7^9
```

```
    * t^5 + 18486 * 7^7 * t^4 + 52512 * 7^5 * t^3 + 79404 * 7^3 * t^2 + 50072 * 7 * t + 16 * 7
    ^2) + p1 * (-6 * 7^13 * t^7 - 207 * 7^11 * t^6 - 3053 * 7^9 * t^5 - 3540 * 7^8 * t^4
    - 16935 * 7^6 * t^3 - 330714 * 7^3 * t^2 - 70359 * 7^2 * t - 6273 * 7 - 2 * 7^2 * t^(-1))
    + 7^14 * t^8 + 10 * 7^13 * t^7 + 1773 * 7^10 * t^6 + 3333 * 7^9 * t^5 + 25701 * 7^7 * t^4
    + 120081 * 7^5 * t^3 + 2322146 * 7^2 * t^2 + 70375 * 7^2 * t + 6273 * 7 + 2 * 7^2 * t^(
    -1)) :
proof7_3_1 := provemodfuncGAMMA0UpETAidBATCH(EP7_3_1, 7, RHS7_3_1, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 27    This means checking up to q^(28).
We find that LHS - RHS is
```

$$O(q^{55})$$

$$proof7_3_1 := [1, -27, O(q^{55})] \quad \textbf{(15)}$$

```
> # GROUP 3, IDENTITY 2: U_A7(t^(-1))
EP7_3_2 := normal(A7/t) :
RHS7_3_2 := expand(p0 * (48 * 7 * t + 48) + p1 * (-48 - 6 * t^(-1)) + 7^2 * t + 48 + 6 * t
    ^(-1)) :
proof7_3_2 := provemodfuncGAMMA0UpETAidBATCH(EP7_3_2, 7, RHS7_3_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$O(q^{34})$$

$$proof7_3_2 := [1, -6, O(q^{34})] \quad \textbf{(16)}$$

```
> # GROUP 3, IDENTITY 3: U_A7(t^(-2))
EP7_3_3 := normal(A7/t^2) :
RHS7_3_3 := expand(p0 * (-80 * 7 * t - 64) + p1 * (78 + 8 * t^(-1)) + 7^3 * t^2 - 8 * 7 * t
    - 78 - 8 * t^(-1)) :
```

```
proof7_3_3 := provemodfuncGAMMA0UpETAidBATCH(EP7_3_3, 7, RHS7_3_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 8    This means checking up to q^(9).
We find that LHS - RHS is
```

$$O(q^{36})$$

$$proof7_3_3 := [1, -8, O(q^{36})] \quad \textbf{(17)}$$

```
> # GROUP 3, IDENTITY 4: U_A7(t^(-3))
  EP7_3_4 := normal(A7/t^3):
  RHS7_3_4 := expand(p0*(12*7^5*t^3+312*7^3*t^2+400*7^2*t+160*7)+p1*(
      -12*7^4*t^2-302*7^2*t-370*7-20*7*t^(-1))+7^5*t^3+108*7^3*t^2
      +2264*7*t+370*7+20*7*t^(-1)):
  proof7_3_4 := provemodfuncGAMMA0UpETAidBATCH(EP7_3_4, 7, RHS7_3_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 12    This means checking up to q^(13).
We find that LHS - RHS is
```

$$O(q^{40})$$

$$proof7_3_4 := [1, -12, O(q^{40})] \quad \textbf{(18)}$$

```
> # GROUP 3, IDENTITY 5: U_A7(t^(-4))

    # Multiply both sides by t^2 to remove negative powers of t on the right, using U_7(t(7*tau)
    ^2*G)=t^2*U_7(G).
  EP7_3_5 := normal(gp2etaprod([49, 8, 7, -8])*A7/t^4):
  RHS7_3_5 := expand(t^2*(p0*(-216*7^4*t^2-2776*7^2*t-1240*7+8*t^(-1))
      +p1*(7^4*t^2+216*7^3*t+2585*7+1084*t^(-1)-t^(-2))-7^7*t^4-167
      *7^3*t^2-240*7^3*t-18087-1084*t^(-1)+t^(-2))):
  proof7_3_5 := provemodfuncGAMMA0UpETAidBATCH(EP7_3_5, 7, RHS7_3_5, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 18    This means checking up to q^(19).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{46})$$

$$proof7_3_5 := [1, -18, \mathrm{O}(q^{46})] \quad \textbf{(19)}$$

```
> # GROUP 3, IDENTITY 6: U_A7(t^(-5))

    # Multiply both sides by t^2 to remove negative powers of t on the right, using U_7(t(7*tau)
    ^2*G)=t^2*U_7(G).
  EP7_3_6 := normal(gp2etaprod([49, 8, 7, -8]) * A7/t^5) :
  RHS7_3_6 := expand(t^2 * (p0 * (6 * 7^9 * t^5 + 76 * 7^7 * t^4 - 18 * 7^7 * t^3 - 580 * 7^4 * t
      ^2 + 10898 * 7^2 * t + 51864 - 104 * t^(-1)) + p1 * (-6 * 7^8 * t^4 - 10 * 7^7 * t^3
      + 129 * 7^5 * t^2 + 446 * 7^3 * t - 10475 * 7 - 6470 * t^(-1) + 13 * t^(-2)) + 7^9 * t^5
      + 62 * 7^7 * t^4 + 54 * 7^6 * t^3 - 871 * 7^4 * t^2 - 302 * 7^3 * t + 73221 + 6470 * t^(
      -1) - 13 * t^(-2))) :
  proof7_3_6 := provemodfuncGAMMA0UpETAidBATCH(EP7_3_6, 7, RHS7_3_6, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 22    This means checking up to q^(23).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{50})$$

$$proof7_3_6 := [1, -22, \mathrm{O}(q^{50})] \quad \textbf{(20)}$$

```
> # GROUP 3, IDENTITY 7: U_A7(t^(-6))

    # Multiply both sides by t^2 to remove negative powers of t on the right, using U_7(t(7*tau)
    ^2*G)=t^2*U_7(G).
  EP7_3_7 := normal(gp2etaprod([49, 8, 7, -8]) * A7/t^6) :
  RHS7_3_7 := expand(t^2 * (p0 * (-12 * 7^11 * t^6 - 440 * 7^9 * t^5 - 100 * 7^9 * t^4 - 104 * 7
```

```
^7*t^3 + 17412*7^4*t^2 + 11112*7^2*t - 33448*7 + 904*t^(-1)) + p1*(12*7
^10*t^5 + 430*7^8*t^4 + 654*7^7*t^3 + 433*7^5*t^2 - 2290*7^4*t - 5427*7
 + 29154*t^(-1) - 113*t^(-2)) - 7^11*t^6 - 124*7^9*t^5 - 3410*7^7*t^4
 - 4386*7^6*t^3 - 1965*7^4*t^2 + 107666*7^2*t + 38893 - 29154*t^(-1) + 113
 *t^(-2))) :
proof7_3_7 := provemodfuncGAMMA0UpETAidBATCH(EP7_3_7, 7, RHS7_3_7, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 25    This means checking up to q^(26).
We find that LHS - RHS is
```

$$O(q^{53})$$

$$proof7_3_7 := [1, -25, O(q^{53})] \quad \textbf{(21)}$$

```
> # GROUP 4, IDENTITY 1: U_B7(q*p0)
  # Since B7*q=1, these identities reduce to U_7(p0*t^(-j)).
  EP7_4_1 := normal(B7*q*p0) :
  RHS7_4_1 := expand(p0*(8*7^7*t^4 + 80*7^5*t^3 + 216*7^3*t^2 + 79*7*t) + p1
      *(7^10*t^5 + 19*7^8*t^4 + 18*7^7*t^3 + 327*7^4*t^2 + 34*7^3*t + 4) + 7^9
      *t^5 + 11*7^7*t^4 + 38*7^5*t^3 + 31*7^3*t^2 + 6*7*t) :
  proof7_4_1 := provemodfuncGAMMA0UpETAidBATCH(EP7_4_1, 7, RHS7_4_1, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof7_4_1 := [1, -15, O(q^{43})] \quad \textbf{(22)}$$

```
> # GROUP 4, IDENTITY 2: U_B7(q*p0*t^(-1))
  EP7_4_2 := normal(B7*q*p0/t) :
  RHS7_4_2 := expand(p0*(-7^3*t^2 - 8*7*t) + p1*(7^2*t + 7) - 7^2*t + 1) :
```

```
proof7_4_2 := provemodfuncGAMMA0UpETAidBATCH(EP7_4_2, 7, RHS7_4_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{34})$$

$$proof7_4_2 := [1, -6, \mathrm{O}(q^{34})] \tag{23}$$

```
> # GROUP 4, IDENTITY 3: U_B7(q*p0*t^(-2))
  EP7_4_3 := normal(B7*q*p0/t^2):
  RHS7_4_3 := expand(p0*(-7^5*t^3-8*7^3*t^2)+p1*(7^4*t^2+7^3*t)-7^4*t^2
      -7^3*t-12):
  proof7_4_3 := provemodfuncGAMMA0UpETAidBATCH(EP7_4_3, 7, RHS7_4_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{37})$$

$$proof7_4_3 := [1, -9, \mathrm{O}(q^{37})] \tag{24}$$

```
> # GROUP 4, IDENTITY 4: U_B7(q*p0*t^(-3))
  EP7_4_4 := normal(B7*q*p0/t^3):
  RHS7_4_4 := expand(p0*(6*7^7*t^4+62*7^5*t^3+27*7^4*t^2+10*7^2*t+1)
      +p1*(-6*7^6*t^3-8*7^5*t^2-3*7^4*t-4*7)+2*7^6*t^3+24*7^4*t^2
      +83*7^2*t+108):
  proof7_4_4 := provemodfuncGAMMA0UpETAidBATCH(EP7_4_4, 7, RHS7_4_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
```

```
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 13    This means checking up to q^(14).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{41})$$

$$proof7_4_4 := [1, -13, \mathrm{O}(q^{41})] \quad \textbf{(25)}$$

```
> # GROUP 4, IDENTITY 5: U_B7(q*p0*t^(-4))
  EP7_4_5 := normal(B7*q*p0/t^4):
  RHS7_4_5 := expand(p0*(-7^9*t^5-50*7^7*t^4-60*7^6*t^3-162*7^4*t^2-60
      *7^2*t-6)+p1*(7^8*t^4+7^8*t^3+54*7^5*t^2+18*7^4*t+24*7)-7^8
      *t^4-25*7^6*t^3-186*7^4*t^2-498*7^2*t-120*7):
  proof7_4_5 := provemodfuncGAMMA0UpETAidBATCH(EP7_4_5,7,RHS7_4_5,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{44})$$

$$proof7_4_5 := [1, -16, \mathrm{O}(q^{44})] \quad \textbf{(26)}$$

```
> # GROUP 4, IDENTITY 6: U_B7(q*p0*t^(-5))
  EP7_4_6 := normal(B7*q*p0/t^5):
  RHS7_4_6 := expand(p0*(-7^11*t^6-8*7^9*t^5+3*7^9*t^4+30*7^7*t^3+81
      *7^5*t^2+30*7^3*t+3*7)+p1*(7^10*t^5+7^9*t^4-3*7^8*t^3-27*7^6
      *t^2-9*7^5*t-12*7^2)-7^10*t^5-7^9*t^4+9*7^7*t^3+93*7^5*t^2
      +249*7^3*t+836*7):
  proof7_4_6 := provemodfuncGAMMA0UpETAidBATCH(EP7_4_6,7,RHS7_4_6,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
```

```
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$O(q^{47})$$

$$proof7_4_6 := [1, -19, O(q^{47})] \quad \textbf{(27)}$$

```
> # GROUP 4, IDENTITY 7: U_B7(q*p0*t^(-6))
  EP7_4_7 := normal(B7*q*p0/t^6):
  RHS7_4_7 := expand(p0*(-7^13*t^7-8*7^11*t^6+2*7^9*t^4+20*7^7*t^3+54
      *7^5*t^2+20*7^3*t+2*7)+p1*(7^12*t^6+7^11*t^5-2*7^8*t^3-18*7
      ^6*t^2-6*7^5*t-8*7^2)-7^12*t^6-7^11*t^5+6*7^7*t^3+62*7^5*t^2
      +166*7^3*t-748*7^2):
  proof7_4_7 := provemodfuncGAMMA0UpETAidBATCH(EP7_4_7,7,RHS7_4_7,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 22    This means checking up to q^(23).
We find that LHS - RHS is
```

$$O(q^{50})$$

$$proof7_4_7 := [1, -22, O(q^{50})] \quad \textbf{(28)}$$

```
> # GROUP 5, IDENTITY 1: U_B7(q*p1*t^(-1))
  # Since B7*q=1, these identities reduce to U_7(p1*t^(-j)).
  EP7_5_1 := expand(B7*q*p1/t):
  RHS7_5_1 := expand(p0*(8*7^8*t^4+80*7^6*t^3+216*7^4*t^2+552*7*t)+p1
      *(7^11*t^5+19*7^9*t^4+18*7^8*t^3+327*7^5*t^2+34*7^4*t+29)+7
      ^10*t^5+11*7^8*t^4+38*7^6*t^3+31*7^4*t^2+41*7*t-1):
  proof7_5_1 := provemodfuncGAMMA0UpETAidBATCH(EP7_5_1,7,RHS7_5_1,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof7_5_1 := [1, -15, O(q^{43})] \quad \textbf{(29)}$$

```
> # GROUP 5, IDENTITY 2: U_B7(q*p1*t^(-2))
  EP7_5_2 := expand(B7*q*p1/t^2):
  RHS7_5_2 := expand(p0*(-8*7^3*t^2-8*7^2*t)+p1*(8*7^2*t+7^2)-7^3*t^2
      -8*7^2*t+11):
  proof7_5_2 := provemodfuncGAMMA0UpETAidBATCH(EP7_5_2, 7, RHS7_5_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 7    This means checking up to q^(8).
We find that LHS - RHS is
```

$$O(q^{35})$$

$$proof7_5_2 := [1, -7, O(q^{35})] \quad \textbf{(30)}$$

```
> # GROUP 5, IDENTITY 3: U_B7(q*p1*t^(-3))
  EP7_5_3 := expand(B7*q*p1/t^3):
  RHS7_5_3 := expand(p0*(-8*7^5*t^3-8*7^4*t^2)+p1*(8*7^4*t^2+7^4*t)-7
      ^5*t^3-8*7^4*t^2-7^4*t-96):
  proof7_5_3 := provemodfuncGAMMA0UpETAidBATCH(EP7_5_3, 7, RHS7_5_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 10    This means checking up to q^(11).
We find that LHS - RHS is
```

$$O(q^{38})$$

$$proof7_5_3 := [1, -10, O(q^{38})] \quad \textbf{(31)}$$

```
> # GROUP 5, IDENTITY 4: U_B7(q*p1*t^(-4))
  EP7_5_4 := expand(B7*q*p1/t^4):
  RHS7_5_4 := expand(p0*(48*7^7*t^4+72*7^6*t^3+216*7^4*t^2+80*7^2*t
      +8)+p1*(-48*7^6*t^3-65*7^5*t^2-24*7^4*t-32*7)-7^7*t^4+16*7^6
```

```
    * t^3 + 199 * 7^4 * t^2 + 664 * 7^2 * t + 740) :
  proof7_5_4 := provemodfuncGAMMA0UpETAidBATCH(EP7_5_4, 7, RHS7_5_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 14    This means checking up to q^(15).
We find that LHS - RHS is
```

$$O(q^{42})$$

$$proof7_5_4 := [1, -14, O(q^{42})] \quad \textbf{(32)}$$

```
> # GROUP 5, IDENTITY 5: U_B7(q*p1*t^(-5))
  EP7_5_5 := expand(B7 * q * p1 / t^5) :
  RHS7_5_5 := expand(p0 * (-8 * 7^9 * t^5 - 64 * 7^8 * t^4 - 80 * 7^7 * t^3 - 216 * 7^5 * t^2
      - 80 * 7^3 * t - 8 * 7) + p1 * (8 * 7^8 * t^4 + 9 * 7^8 * t^3 + 72 * 7^6 * t^2 + 24 * 7^5 * t
      + 32 * 7^2) - 7^9 * t^5 - 8 * 7^8 * t^4 - 31 * 7^7 * t^3 - 248 * 7^5 * t^2 - 664 * 7^3 * t
      - 740 * 7) :
  proof7_5_5 := provemodfuncGAMMA0UpETAidBATCH(EP7_5_5, 7, RHS7_5_5, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 17    This means checking up to q^(18).
We find that LHS - RHS is
```

$$O(q^{45})$$

$$proof7_5_5 := [1, -17, O(q^{45})] \quad \textbf{(33)}$$

```
> # GROUP 5, IDENTITY 6: U_B7(q*p1*t^(-6))
  EP7_5_6 := expand(B7 * q * p1 / t^6) :
  RHS7_5_6 := expand(p0 * (-8 * 7^11 * t^6 - 8 * 7^10 * t^5 + 40 * 7^9 * t^4 + 400 * 7^7 * t^3
      + 1080 * 7^5 * t^2 + 400 * 7^3 * t + 40 * 7) + p1 * (8 * 7^10 * t^5 + 7^10 * t^4 - 40 * 7^8
      * t^3 - 360 * 7^6 * t^2 - 120 * 7^5 * t - 160 * 7^2) - 7^11 * t^6 - 8 * 7^10 * t^5 - 7^10 * t^4
      + 120 * 7^7 * t^3 + 1240 * 7^5 * t^2 + 3320 * 7^3 * t + 4720 * 7) :
  proof7_5_6 := provemodfuncGAMMA0UpETAidBATCH(EP7_5_6, 7, RHS7_5_6, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 20    This means checking up to q^(21).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{48})$$

$$proof7_5_6 := [1, -20, \mathrm{O}(q^{48})] \quad \textbf{(34)}$$

```
> # GROUP 5, IDENTITY 7: U_B7(q*p1*t^(-7))
  # Multiply both sides by t to remove t^(-1) on the right, using U_7(t(7*tau)*G)=t*U_7(G).
  EP7_5_7 := expand(gp2etaprod([49, 4, 7, -4]) * B7 * q * p1 / t^7) :
  RHS7_5_7 := expand(t* (p0 * (-8 * 7^13 * t^7 - 8 * 7^12 * t^6 - 136 * 7^9 * t^4 - 1360 * 7^7
      * t^3 - 3672 * 7^5 * t^2 - 1360 * 7^3 * t - 136 * 7) + p1 * (8 * 7^12 * t^6 + 6 * 7^11 * t^5
      - 19 * 7^9 * t^4 + 818 * 7^7 * t^3 + 1167 * 7^6 * t^2 + 2798 * 7^4 * t + 11 * 7^4 + t^(
      -1)) + 6 * 7^13 * t^7 + 20 * 7^12 * t^6 + 272 * 7^10 * t^5 + 1893 * 7^8 * t^4 + 435 * 7^7
      * t^3 - 2953 * 7^5 * t^2 - 10620 * 7^3 * t - 27029 * 7)) :
  proof7_5_7 := provemodfuncGAMMA0UpETAidBATCH(EP7_5_7, 7, RHS7_5_7, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 25    This means checking up to q^(26).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{53})$$

$$proof7_5_7 := [1, -25, \mathrm{O}(q^{53})] \quad \textbf{(35)}$$

```
> # GROUP 6, IDENTITY 1: U_B7(q)=U_7(1)=1, directly from the definition of U_7.

> # GROUP 6, IDENTITY 2: U_B7(q*t^(-1))
  EP7_6_2 := normal(B7 * q / t) :
  RHS7_6_2 := -7 * t - 4 :
  proof7_6_2 := provemodfuncGAMMA0UpETAidBATCH(EP7_6_2, 7, RHS7_6_2, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
```

```
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 3    This means checking up to q^(4).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{31})$$

$$proof7_6_2 := [1, -3, \mathrm{O}(q^{31})] \quad \textbf{(36)}$$

```
> # GROUP 6, IDENTITY 3: U_B7(q*t^(-2))
  EP7_6_3 := normal(B7*q/t^2):
  RHS7_6_3 := -7^3*t^2+20:
  proof7_6_3 := provemodfuncGAMMA0UpETAidBATCH(EP7_6_3,7,RHS7_6_3,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{34})$$

$$proof7_6_3 := [1, -6, \mathrm{O}(q^{34})] \quad \textbf{(37)}$$

```
> # GROUP 6, IDENTITY 4: U_B7(q*t^(-3))
  EP7_6_4 := normal(B7*q/t^3):
  RHS7_6_4 := -7^5*t^3-88:
  proof7_6_4 := provemodfuncGAMMA0UpETAidBATCH(EP7_6_4,7,RHS7_6_4,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$O(q^{37})$$

$$proof7_6_4 := [1, -9, O(q^{37})] \quad \textbf{(38)}$$

```
> # GROUP 6, IDENTITY 5: U_B7(q*t^(-4))
  EP7_6_5 := normal(B7*q/t^4):
  RHS7_6_5 := -7^7*t^4+260:
  proof7_6_5 := provemodfuncGAMMA0UpETAidBATCH(EP7_6_5,7,RHS7_6_5,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 13    This means checking up to q^(14).
We find that LHS - RHS is
```

$$O(q^{41})$$

$$proof7_6_5 := [1, -13, O(q^{41})] \quad \textbf{(39)}$$

```
> # GROUP 6, IDENTITY 6: U_B7(q*t^(-5))
  EP7_6_6 := normal(B7*q/t^5):
  RHS7_6_6 := -7^9*t^5+68*7:
  proof7_6_6 := provemodfuncGAMMA0UpETAidBATCH(EP7_6_6,7,RHS7_6_6,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$O(q^{44})$$

$$proof7_6_6 := [1, -16, O(q^{44})] \quad \textbf{(40)}$$

```
> # GROUP 6, IDENTITY 7: U_B7(q*t^(-6))
  EP7_6_7 := normal(B7*q/t^6):
  RHS7_6_7 := -7^11*t^6-2392*7:
  proof7_6_7 := provemodfuncGAMMA0UpETAidBATCH(EP7_6_7,7,RHS7_6_7,14);
```

```
*** There were NO errors.
```

```
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{47})$$

$$proof7_6_7 := [1, -19, \mathrm{O}(q^{47})] \quad \textbf{(41)}$$

```
>
```